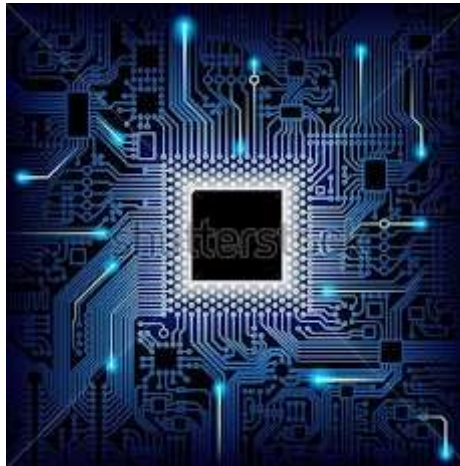


Quiz Test, Course Code: CSE - 2104, Course Title: Sessional Based on CSE - 2103, Time: 20 Minutes, Full Marks: 20

Total points **17/20** ?

0 of 0 points

General Instructions: Fill the roll and email very carefully, otherwise your answer script will not be evaluated. There is no alternative questions & no negative marks for wrong answers.



Basic Information

0 of 0 points

Please, write the roll number carefully, otherwise your answer script will not be evaluated.

Roll? *

2203152

Questions :

17 of 20 points



Instruction: Answer all the questions. There is no negative marks for wrong answers. The

figures in the right margin indicate full marks for the respective question.

- ✓ How many FFs are required for a counter that will count 0 to 64_{10} ? (Write only decimal/numeric value in answer, otherwise your answer will not be evaluated.) *1/1

7



- ✗ What is the frequency of the output of the 4th (starting from MSB flip flop) flip-flop (FF) of an asynchronous counter when the input clock frequency is 2048 kHz? Assume that the counter can count 0 to 31_{10} . *0/1

☐ 2048 kHz

☐ 1024 kHz

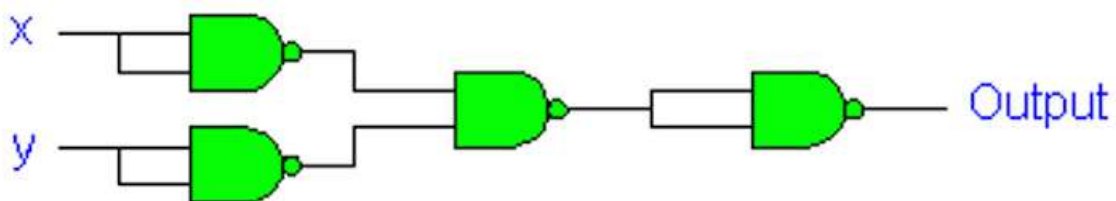
☐ 64 kHz

☒ 128 kHz



- ✓ Which gate is the following circuit equivalent to? *

2/2



☐ AND

☐ OR

☐ NAND

☒ NOR



✓ Karnaugh map (K-map) technique provides a systematic method for simplifying ----- *1/1

☐ logic gates

☐ none of these

☐ multiplexers

☒ Boolean expressions ✓

✓ What is the MOD number of the counter that will count 0 to 307_{10} ? (Write only decimal value in answer, otherwise your answer will not be evaluated.) *1/1

308 ✓

✓ If a binary counter starts at 000, what will be its output state after 91 pulses? Suppose, all the flip-flops are functioning. (Write only binary value in answer, otherwise your answer will not be evaluated.) *1/1

011 ✓

✓ Usually D flip-flops are used to design for_____ (Check all that can apply) *1/1

☐ Register and subtractor

☐ Adder and counter

☐ none of these

☒ Counter and register ✓



✓ Are these two equations equal? *

1/1

$$A\bar{B}C + \overline{ABC} = \bar{B}(AC + \bar{A}\bar{C})$$

☐ Yes

☒ No



✓ Given $X = A'B + (C' + E)(D + F')$, use de Morgan's theorem to find X' (Complement of X).

*2/2

☒ $ACE' + AD'F + B'CE' + B'D'F$



☐ $ACE' + BCE' + D'F$

☐ $(A + B')(CE'D'F)$

☐ $A + B + CE'D'F$



✗ Determine the expression for the following K- map. (Check all that can apply) *0/2

	$\bar{C}$	C
$\bar{A}\bar{B}$	1	1
$\bar{A}B$	1	0
AB	1	X
$A\bar{B}$	1	X

☒ $B' + C'$



☐ $(BC)'$

☐ $A + B' + C'$

☐ C'

✓ Which of the following option(s) is/are the correct statement(s)? (Check all that can apply) *1/1

☒ IC number of 1 input NOT gate is 7404



☒ IC number of 2 input AND gate is 7408



☐ IC number of 2 input NAND gate is 7408

☐ IC number of 2 input OR gate is 7404



✓ What FF outputs should be connected to the clearing NAND gate to form a MOD 13 counter? Assume, Assume that the bit count order is ABCD where A is MSB and D is LSB bit. *1/1

☒ A, B, D ✓

☐ B, C, D

☐ A, C, D

☐ A, B, C

✓ Combinational circuit has no _____. * 1/1

☐ Basic gates

☐ none of these

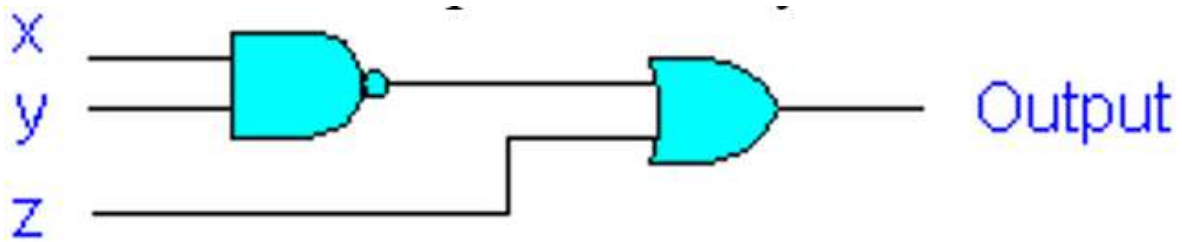
☒ Memory elements ✓

☐ Universal Gates



✓ What function is implemented by the circuit shown in the following figure? (Check all that can apply)

*2/2



☐ $(x' + y')z$

☒ $(xyz)'$

✓

☒ $x' + y' + z$

✓

☐ $x'y'z$

✓ At least, how many byte(s) are needed to represent $(51)_{10}$ in binary? *

1/1

☒ 1

✓

☐ 7

☐ 0.7

☐ 6



✓ If the present state is 1 and the next state is 0, what will be the control inputs of JK flip flop, assuming that JK flip flop is made by NOR gate? *1/1
(Check all that can apply).

☒ J = 0, K = 1 ✓

☐ J = 1, K = 0

☐ J = 0, K = 0

☒ J = 1, K = 1 ✓

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